

Geometry, Entropy, and Operator Learning with Tunable Kernels

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Abstract

We study a two-dial family of kernels

$$k_{\theta,\sigma}(y \mid x) \propto w_{\theta}(\|x - y\|) g_{\sigma}(y - x),$$

whose product form covers any radial weight w_{θ} and centred residual density g_{σ} that satisfy minimal smoothness and moment conditions. To give the theory a concrete footing—and to obtain closed-form bias constants in our finite-sample analysis—we instantiate this family with a data-adaptive *local-linear* construction, but all results hold for the broader class.

For every observable f we compute the one-step cumulant-generating function (CGF) and show that its quadratic coefficient $\mathcal{I}_{\theta,\sigma}[f]$ serves simultaneously as (i) a *Fisher load*, (ii) $\frac{2}{\theta^2}$ times the synthetic Ricci curvature $\Gamma_2^{(\theta,\sigma)}(f)$, and (iii) $-2 \partial_{\sigma}$ of the one-step Shannon entropy. This *Fisher-curvature-entropy bridge* yields a single quadratic objective whose minimiser in the dial plane, (θ^*, σ^*) —the *little-rose*—optimises curvature bias, entropy production, and operator-learning risk at once. We prove

$$\nabla_{(\theta,\sigma)} \text{KL}(\mu_1 \parallel k_{\theta,\sigma} \# \mu_0) = -\frac{1}{2} \nabla_{(\theta,\sigma)} \mathcal{I}_{\theta,\sigma},$$

so steepest KL descent converges to the little-rose with no hyper-parameters. A data-driven Goldenshluger–Lepski selector recovers $(\hat{\theta}_{\star}, \hat{\sigma}_{\star})$ from finite samples, and we give non-asymptotic risk bounds for both Perron–Frobenius and Koopman estimates.

The resulting dial matrices $(\Theta_{ij}, \Sigma_{ij})$ form a directed, doubly-weighted graph, opening avenues toward discrete curvature flows, entropy production on cycles, and multiparameter persistent homology—highlighting a promising intersection of statistical dynamics, information geometry, and topological data analysis.

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1 Introduction

1.1 Why tune local kernels?

1.2 Key contributions

- Closed-form bridge: CGF curvature $\rightarrow$ Fisher load $\rightarrow$ curvature & entropy.
- Single KL optimisation (little-rose) jointly tunes **two** dials (θ, σ) .
- Finite-sample Goldenshluger-Lepski selector with risk guarantees.

1.3 Organisation of the paper

2 Rose kernels and dials

2.1 Product-form kernels and minimal assumptions

$$z := y - x, \quad r := \|z\|.$$

(W1) **(Radial weight)** $w : [0, \infty) \rightarrow [0, \infty)$ is C^1 , radial, $\int_{\mathbb{R}^d} w(\|z\|) dz = 1$, $\int \|z\|^2 w(\|z\|) dz < \infty$, and $w \in L^2(\mathbb{R}^d)$.

(W2) **(Centred residual)** $g : \mathbb{R}^d \rightarrow [0, \infty)$ is C^1 , $\int g(z) dz = 1$, $\int z g(z) dz = 0$, $\int z z^\top g(z) dz < \infty$, and $\nabla g \in L^2(\mathbb{R}^d)$.

Definition 2.1 (Rose kernel). For $(\theta, \sigma) \in (0, \infty)^2$ set

$$w_\theta(r) := \theta^{-d} w(r/\theta), \quad g_\sigma(z) := \sigma^{-d} g(z/\sigma), \quad k_{\theta,\sigma}(y | x) := \frac{w_\theta(\|x - y\|) g_\sigma(y - x)}{\int_X w_\theta(\|x - y'\|) g_\sigma(y' - x) dy'}$$

Lemma 2.1 (Normalisation and basic bounds). *Under Assumptions (W1)–(W2) the expression in Definition 2.1 defines a measurable Markov transition density $k_{\theta,\sigma}(y | x)$:*

$$0 \leq k_{\theta,\sigma}(y | x) \leq \frac{w_\theta(\|x - y\|) g_\sigma(y - x)}{\inf_x Z_{\theta,\sigma}(x)}, \quad \int_X k_{\theta,\sigma}(y | x) d\mu(y) = 1,$$

where $Z_{\theta,\sigma}(x)$ is the denominator in Definition 2.1 and satisfies $0 < \inf_x Z_{\theta,\sigma}(x) \leq \sup_x Z_{\theta,\sigma}(x) < \infty$.

(K1) **(Markov)** $k_{\theta,\sigma}(y | x) \geq 0$ and $\int_X k_{\theta,\sigma}(y | x) dy = 1$.

(K2) **(Score in L^2)** $S_{\theta,\sigma}(y | x) := \nabla_y \log k_{\theta,\sigma}(y | x) \in L^2(dy dx)$.

(K3) **(Second-moment scaling)** $\int_X \|y - x\|^2 k_{\theta,\sigma}(y | x) dy = O(\theta^2)$ as $\theta \downarrow 0$.

(K4) **(C^1 in dials)** $\partial_\theta \log k_{\theta,\sigma}, \partial_\sigma \log k_{\theta,\sigma} \in L^2(dy dx)$.

2.2 Local-linear residual construction (canonical example)

2.3 Scaled kernels and notation for the rest of the paper

2.4 Neutral (reversible) Rose operator

- Define $k^{\natural}(y | x) = \frac{1}{2}[k(y | x) + k(x | y)]$.
- Self-adjoint $\mathcal{R}^{\natural}$; serves as equilibrium baseline.

3 Population theory

(K0)–(K2) as in Sec. 3.0 — Markov, score L^2 , second-moment scaling.

3.1 CGF $\Rightarrow$ Fisher $\Rightarrow$ Landau

Definition 3.1 (One-step CGF).

Definition 3.2 (Symmetric Fisher load). Given a Rose kernel $k_{\theta,\sigma}(y | x)$ with score $S_{\theta,\sigma}(y | x) := \nabla_y \log k_{\theta,\sigma}(y | x)$ and anchor/target measures μ_0, μ_1 , define for $f \in W^{1,2}(X, \mu)$

$$\mathcal{I}_{\theta,\sigma}[f] := \iint_{X \times X} \langle \nabla f(y), S_{\theta,\sigma}(y | x) \rangle^2 d\mu_0(x) d\mu_1(y).$$

Similarly, for a density $\rho \ll \mu$,

$$\mathcal{I}_{\theta,\sigma}[\rho] := \iint_{X \times X} \langle \nabla \log \rho(y), S_{\theta,\sigma}(y | x) \rangle^2 d\mu_0(x) d\mu_1(y).$$

Proposition 3.1 (CGF–Fisher correspondence). *For every $(\theta, \sigma) \in (0, \infty)^2$ and every centred observable $f \in W^{1,2}(X, \mu)$,*

$$\Lambda_{\theta,\sigma}(\lambda; f) = \frac{\lambda^2}{2} \mathcal{I}_{\theta,\sigma}[f] + O(\lambda^3), \quad \lambda \rightarrow 0,$$

where $\mathcal{I}_{\theta,\sigma}[f] = \iint \langle \nabla f(y), S_{\theta,\sigma}(y | x) \rangle^2 d\mu_0(x) d\mu_1(y)$.

Remark 3.1 (Legendre dual = Landau stiffness). A two-line box reminding readers that the Legendre transform of Λ yields a quadratic Landau free energy.

3.2 Dirichlet energy as common source

Definition 3.3 (Symmetrised kernel and Dirichlet form). Let $k_{\theta,\sigma}^{\natural}(y | x) := \frac{1}{2}(k_{\theta,\sigma}(y | x) + k_{\theta,\sigma}(x | y))$ and let $\mu_{\natural}$ be any reference measure absolutely continuous w.r.t. both μ_0 and μ_1 (often $\mu_{\natural} = \frac{1}{2}(\mu_0 + \mu_1)$). Define the associated Dirichlet form

$$\mathcal{E}_{\theta,\sigma}(f) := \frac{1}{2} \iint_{X \times X} (f(y) - f(x))^2 k_{\theta,\sigma}^{\natural}(y | x) d\mu_{\natural}(x) d\mu_{\natural}(y).$$

Lemma 3.1 (Fisher load from Dirichlet energy). *Under Assumptions (K1)–(K3), there exists a dial-dependent coefficient $c_{\theta,\sigma} > 0$ such that*

$$\mathcal{I}_{\theta,\sigma}[f] = c_{\theta,\sigma} \mathcal{E}_{\theta,\sigma}(f) + o(1) \quad (\theta \downarrow 0),$$

with $c_{\theta,\sigma} = 4/\theta^2 + o(\theta^{-2})$ in the isotropic small-locality regime.

Remark: Reversible baseline and operator split

3.3 Fisher–curvature–entropy bridge

Lemma 3.2 (Iterated carré–du–champ). *Under Assumptions (K1)–(K3), for every $f \in W^{1,2}(X, \mu)$,*

$$\Gamma_2^{(\theta,\sigma)}(f) = \frac{\theta^2}{2} \mathcal{I}_{\theta,\sigma}[f] + o(\theta^2), \quad \theta \downarrow 0.$$

Lemma 3.3 (Discrete de Bruijn Identity). *Fix $\theta > 0$. For every density $\rho \ll \mu$ with $\mathcal{I}_{\theta,\sigma}[\rho] < \infty$,*

$$\frac{d}{d\sigma} H(R_{\theta,\sigma}^\# \rho) = -\frac{1}{2} \mathcal{I}_{\theta,\sigma}[\rho], \quad \sigma > 0.$$

Theorem 3.1 (Fisher–curvature–entropy bridge). *With Assumptions (K1)–(K3),*

$$\boxed{\mathcal{I}_{\theta,\sigma} = \frac{2}{\theta^2} \Gamma_2^{(\theta,\sigma)} = -2 \frac{d}{d\sigma} H(R_{\theta,\sigma}^\# \rho)} + o(1).$$

Corollary 3.1 (Discrete fluctuation–dissipation). *Combine Lemma 3.2 and Lemma 3.3. In the small–locality regime $\theta \downarrow 0$, the response coefficient obtained by differentiating the one–step entropy $H(R_{\theta,\sigma}^\# \rho)$ w.r.t. the noise dial σ equals (up to the universal factor $2/\theta^2$) the curvature prefactor $\Gamma_2^{(\theta,\sigma)}(f)$, i.e.*

$$\frac{d}{d\sigma} H(R_{\theta,\sigma}^\# \rho) = -\frac{1}{\theta^2} \Gamma_2^{(\theta,\sigma)}(f) + o(1).$$

Thus fluctuations (Fisher load) determine dissipation (entropy production) in exact analogy with the Onsager reciprocal relations.

3.4 Discrete Onsager–Machlup functional

Corollary 3.2 (Discrete OM action). *Let $\mathbf{x} = (x_0, \dots, x_n)$ be a trajectory sampled every h and let $k_{\theta,\sigma}$ satisfy (K0)–(K2). With local drift m_θ and quadratic forms $\Gamma_t^2 := \Gamma_2^{(\theta,\sigma)}(f_t)$, $\dot{H}_t := \partial_\sigma H(R_{\theta,\sigma}^\# \rho_t)$,*

$$\mathcal{A}_{\theta,\sigma}[\mathbf{x}] = \frac{1}{\theta^2} \sum_t \Gamma_t^2 - \frac{1}{\sigma^2} \sum_t \dot{H}_t + o(1)$$

as $(\theta, \sigma) \downarrow 0$. If $k = k^\natural$ and $\sigma^2 = 2Dh$ with $h \rightarrow 0$, this reduces to the classical continuous Onsager–Machlup functional.

Remark 3.2 (Data-driven OM). Any consistent drift estimate (e.g. LWLS–1) may replace m_θ without changing the quadratic structure of $\mathcal{A}_{\theta,\sigma}[\mathbf{x}]$.

3.5 Deterministic limit and least–action equivalence

Remark 3.3 (Classical Lagrangian limit). In the joint limit $\theta, h \rightarrow 0$ with $\sigma^2 = 2Dh$, the discrete action density converges to $\frac{1}{4D} \int_0^T \|\dot{x}(t) - b(x(t))\|^2 dt$, the standard Onsager–Machlup Lagrangian for diffusions.

Theorem 3.2 (KL–least–action equivalence). *For every Rose kernel $k_{\theta,\sigma}$ let*

$$J(\theta, \sigma) := \text{KL}(\mu_1 \parallel k_{\theta,\sigma} \# \mu_0), \quad \mathcal{A}_{\theta,\sigma}^{\text{pop}} := \mathbb{E}_{\mu_0}[\mathcal{A}_{\theta,\sigma}[X_{0:n}]]$$

denote, respectively, the population Kullback–Leibler functional and the population Onsager–Machlup action introduced in Corollary 3.2. Under Assumptions (K0)–(K3)

$$\boxed{\nabla_{(\theta,\sigma)} J = \nabla_{(\theta,\sigma)} \mathcal{A}_{\theta,\sigma}^{\text{pop}}} \quad \text{for all } (\theta, \sigma) \in (0, \infty)^2.$$

Remark 3.4. The pair

$$(\theta^*, \sigma^*) := \arg \min_{(\theta, \sigma)} J(\theta, \sigma) = \arg \min_{(\theta, \sigma)} \mathcal{A}_{\theta, \sigma}^{\text{pop}}$$

maximises the likelihood of the observed path ensemble within the two-dial Rose family; equivalently, it minimises the expected Onsager–Machlup action. Hence trajectories generated under the little-rose kernel are the most probable (in the maximum-likelihood sense) among all trajectories that can be produced by Rose kernels.

3.6 Introducing the dial plane

(K4) C^1 regularity in dials. $(\theta, \sigma) \mapsto \log k_{\theta, \sigma}$ is C^1 and $\partial_\theta \log k_{\theta, \sigma}, \partial_\sigma \log k_{\theta, \sigma} \in L^2(d\mu_0 d\mu_1)$.

3.7 Edge-wise dial oracle

Oracle axioms

- (A1) Reversibility consistency.** If $k(y | x) = k(x | y)$ then $\mathcal{O}(x_i, x_j) = \mathcal{O}(x_j, x_i)$.
- (A2) Permutation equivariance.** $\mathcal{O}(\pi(i), \pi(j)) = \mathcal{O}(x_i, x_j)$ for any permutation π .
- (A3) Strict-convex minimal risk.** $\mathcal{O}(x_i, x_j) = \arg \min_{(\theta, \sigma)} \mathcal{J}_{ij}(\theta, \sigma)$ with $\mathcal{J}$ strictly convex.
- (A4) Positive empirical mass.** Empirical anchor weights satisfy $\inf_{i \in V} \hat{\mu}_0(i) > 0$.

Definition 3.4 (Edge KL functional). Given observed transitions from anchor x_i , let $\hat{k}_{ij}$ denote the empirical conditional probability that a step from x_i lands in x_j . Define the edge-wise KL at dials (θ, σ) :

$$\mathcal{J}_{ij}(\theta, \sigma) := \sum_j \hat{k}_{ij} \log \frac{\hat{k}_{ij}}{k_{\theta, \sigma}(x_j | x_i)}.$$

(For continuous X , replace the sum by an integral against the empirical conditional measure $\hat{k}_i$.)

Proposition 3.2 (Edge-wise KL gradient). *For every ordered pair (x_i, x_j) and all (θ, σ) ,*

$$\nabla_{(\theta, \sigma)} \mathcal{J}_{ij}(\theta, \sigma) = -\frac{1}{2} \nabla_{(\theta, \sigma)} \mathcal{I}_{ij}(\theta, \sigma).$$

Consequently the oracle dials $(\theta_{ij}^, \sigma_{ij}^*)$ maximise the Fisher load and minimise the edge-wise KL.*

Definition 3.5 (Little-rose operator). Assign each edge its oracle dials $(\theta_{ij}^*, \sigma_{ij}^*)$ and denote by $\mathcal{R}_*^{b/\#}$ the resulting pull-back and push-forward operators. We call this pair the *little-rose operator*.

3.8 Global KL gradient corollary

Corollary 3.3 (Global KL gradient). *Summing Proposition 3.2 over all edges with weights $\mu_0(i)$ yields*

$$\nabla_{(\theta, \sigma)} \text{KL}(\mu_1 \parallel k_{\theta, \sigma} \# \mu_0) = -\frac{1}{2} \nabla_{(\theta, \sigma)} \mathcal{I}_{\theta, \sigma},$$

hence the little-rose operator is the unique minimiser of KL and maximiser of the Fisher load.

4 Finite-sample dial selection

4.1 Goldenshluger–Lepski selector

Lemma 4.1 (Bias–variance decomposition). *For either PF or Koopman risk,*

$$\mathbb{E} \widehat{\mathcal{R}}^\square(\theta, \sigma) = \text{Bias}_\square^2(\theta, \sigma) + \text{Var}_\square(\theta, \sigma),$$

where the bias term is the squared L^2 distance between the population Rose operator at (θ, σ) and the “oracle” operator that best fits the data class, and the variance term scales as

$$\text{Var}_\square(\theta, \sigma) \lesssim \frac{\text{trace}(\Sigma_{\theta, \sigma}^\square)}{n_{\text{eff}}(\theta)},$$

with an effective sample size $n_{\text{eff}}(\theta)$ determined by the local weight w_θ (see Appendix ??).

Theorem 4.1 (GL selector: non-asymptotic risk). *Let $\widehat{\theta}, \widehat{\sigma}$ be chosen by the GL rule with penalty parameter α . Then with probability at least $1 - \delta$,*

$$\mathcal{R}(\widehat{\theta}, \widehat{\sigma}) \leq C_\alpha \inf_{(\theta, \sigma) \in \mathcal{G}} \mathcal{R}(\theta, \sigma) + C \frac{\log(1/\delta)}{n},$$

where $\mathcal{R}$ denotes either PF or Koopman L^2 -risk and C_α, C are explicit constants (see Appendix ??).

Definition 4.1 (Sample risks). Let $\widehat{R}_{\theta, \sigma}^\#$ and $\widehat{R}_{\theta, \sigma}^b$ denote the empirical PF and Koopman estimators built from the sample. Given a test class $\mathcal{F} \subset L^2(X, \mu)$ and $\mathcal{P} \subset L^1(X, \mu)$ define

$$\widehat{\mathcal{R}}^{\text{PF}}(\theta, \sigma) := \sup_{\rho \in \mathcal{P}} \|\widehat{R}_{\theta, \sigma}^\# \rho - R_{\theta, \sigma}^\# \rho\|_{L^2}^2, \quad \widehat{\mathcal{R}}^{\text{K}}(\theta, \sigma) := \sup_{f \in \mathcal{F}} \|\widehat{R}_{\theta, \sigma}^b f - R_{\theta, \sigma}^b f\|_{L^2}^2.$$

5 Directed curvature and reversibility diagnostics

5.1 Curvature proxy on the dial graph

5.2 Irreversibility measure via antisymmetric current

Definition 5.1 (Antisymmetric weight & current). For a Rose kernel $k_{\theta, \sigma}$ define the log–ratio antisymmetry

$$A_{ij} := \frac{1}{2} \left(\log k_{\theta, \sigma}(x_j | x_i) - \log k_{\theta, \sigma}(x_i | x_j) \right)$$

Given a reference mass $\pi_i > 0$ on vertices (e.g. stationary or empirical mass), the probability current is

$$J_{ij} := \pi_i k_{\theta, \sigma}(x_j | x_i) - \pi_j k_{\theta, \sigma}(x_i | x_j),$$

satisfying $J_{ij} = -J_{ji}$. Detailed balance holds iff all $A_{ij} = 0$ (equivalently $J_{ij} = 0$).

Lemma 5.1 (Graph Helmholtz split). *Let J be an antisymmetric edge flow on a finite connected graph (V, E) with vertex weights π . Then there exists a potential $\phi : V \rightarrow \mathbb{R}$ and a cycle flow $h \in \ker B^\top$ (where B is the weighted incidence matrix) such that*

$$J_{ij} = \pi_i k_{ij}^h (\phi_j - \phi_i) + h_{ij},$$

with $\sum_j \pi_i k_{ij}^h (\phi_j - \phi_i) = 0$ at stationarity. The decomposition is unique up to an additive constant in ϕ . Moreover $h = 0$ iff detailed balance holds.

Remark 5.1 (Loschmidt symmetry and the Fisher clock). Setting every antisymmetric weight to zero, $A_{ij} = 0$, collapses the kernel to its neutral form k^h and makes the Rose operator exactly time-reversal symmetric, matching Loschmidt’s objection to irreversible entropy growth.

Turning on any $A \neq 0$ injects a directed cycle curvature $F = \sum_{\text{cycle}} A$, and the Fisher–curvature–entropy bridge (Thm. 3.1) yields

$$F = \frac{1}{2} \mathcal{I}_{\theta, \sigma} = \text{entropy production per step.}$$

A non-zero curvature therefore acts as an intrinsic *Fisher clock*—time emerges only once detailed balance is broken. In the Loschmidt-symmetric limit $A \rightarrow 0$ the clock halts and the macroscopic arrow of time disappears.

6 Discussion and Outlook

6.1 Key implications

(I1) Unified diagnostic of fluctuation strength

The Fisher load $\mathcal{I}_{\theta, \sigma}$ is *simultaneously* the variance coefficient, the synthetic–curvature prefactor, and the entropy–production rate. A single dial pair (θ, σ) governs them all.

(I2) Automatic bias–variance balance

Steepest descent of $\text{KL}(\mu_1 \parallel k_{\theta, \sigma} \# \mu_0)$ is steepest *ascent* of $\mathcal{I}_{\theta, \sigma}$, so the statistically most informative kernel is found with no cross-validation.

(I3) One-step Schrödinger bridge

The little-rose kernel realises the discrete entropic transport $\mu_0 \rightarrow \mu_1$, linking operator learning to optimal-transport theory.

(I4) Landau lens on data

The Legendre transform of the CGF equips any observable with a Landau-type free-energy landscape whose stiffness is data-computable.

(I5) Lag–dimension selection. Remark ?? suggests using the little-rose Fisher load as a principled criterion for choosing delay-embedding parameters.

6.2 Future directions

(F1) Directed discrete curvature

Dial matrices $(\Theta_{ij}, \Sigma_{ij})$ define a doubly-weighted digraph, opening analytic work on Ollivier/Bakry–Émery curvature for non-reversible kernels.

(F2) Multiparameter persistent homology

The two weights furnish a natural bifiltration, inviting multi-parameter TDA of dynamical data sets.

6.3 Limitations

6.4 Conclusion

A Proof details for Sec. 3

B GL selector derivations

C Additional experiments

D Dial-graph topology (exploratory)